

MOUNTAINS OF THE MOON UNIVERSITY

FACULTY OF SCIENCE TECHNOLOGY AND INNOVATION

GROUP COURSE WORK

GROUP 8

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QUESTIONS

- a). What is computer recycling or electronic recycling.
- b). As of Thursday, 10 JUNE 2021, Government of Uganda under the National Environment Management Authority (NEMA) together with the National Enterprise Corporation (NEC) launched the first National e-waste Management center to promote safe and proper disposal of all outdated and defunct electronic equipment in the country Explain the effects of E-waste on the environment.
- c). Differentiate between green computing and cloud computing.
- d). Give five benefits of adopting to cloud computing.

a). Computer recycling, also known as electronic recycling or e-waste recycling, is the process of recovering valuable materials and components from old or obsolete electronic devices and computers. This environmentally responsible practice aims to divert electronic waste from landfills and incinerators, reducing the negative environmental impact of electronic waste disposal.

The recycling process typically involves the following steps:

Collection: Electronic devices, such as computers, laptops, smartphones, and other electronic equipment, are collected from consumers, businesses, or electronic waste disposal centers.

Sorting: The collected electronics are sorted based on their type and condition. Reusable and recyclable components are separated from non-recyclable materials.

Dismantling: Electronics are disassembled into their individual components and parts. Valuable materials like metals (e.g., copper, aluminum, gold) and plastics are extracted.

Refurbishment and Resale: Some electronics, if in good working condition, can be refurbished and resold in the second-hand market.

Recycling: Non-reusable parts and materials are sent for recycling. Valuable metals are extracted, and plastics may be processed for reuse.

Safe Disposal: Any hazardous materials, such as batteries or toxic components, are disposed of safely and in accordance with environmental regulations.

Conclusion

Computer and electronic recycling not only conserves natural resources but also reduces the energy and environmental costs associated with manufacturing new electronic products. It also helps prevent the release of harmful substances found in electronic waste when improperly disposed off, such as lead, mercury, and cadmium. Many countries and regions have established regulations and recycling programs to encourage proper e-waste disposal and recycling.

b) E-waste, or electronic waste, refers to discarded electronic devices and equipment that have reached the end of their useful life. This category includes items like old computers, smartphones, televisions, and other electronic gadgets. E-waste can be harmful to the environment if not properly managed due to the presence of hazardous materials like lead, mercury, and cadmium in some electronic components. Recycling and

responsible disposal of e-waste are essential to mitigate its environmental impact and recover valuable resources. Electronic waste has negative significance on the environment as follow:

- ❖ **Impact on Air Quality;** One of the most common effects of E-waste on the environment is air pollution electronic-waste can pollute the air when improperly shredded, melted, or burned. These practices release dust particles or toxins, such as dioxins, into the environment and cause air pollution. The chemicals released while burning e-waste can travel thousands of miles and cause serious health concerns for all living things. E-waste of little value is often burned, but burning also serves as a way to get valuable metals from electronics, like copper. However, this burning exposes those in the area to the toxins in the air because it quickly and easily spreads across the region and beyond. Whens these toxins are released into the air, they can travel for miles. This can lead to multiple people being forced to breathe in contaminated air, leading to further problems such as respiratory concerns. Chronic diseases and cancers are at a higher risk of occurring as a result electronic waste
- ❖ **Climate change:** It is also worth considering the effect that e- waste has on climate change, every device ever produce has a carbon foot

print and it is contributing to human made global warming. It contributes to climate change by emitting greenhouse gasses and consuming energy

- ❖ **Impact on biodiversity:** The consequence of improper e-waste disposal in the land fill or other non-dumping sites poses serious threats to the environment and can pollute ecosystems for generation to come. When electronic waste is thrown away in the , their toxic materials seep into ground water affecting both land and sea water animals .acidification from the leachate of the heavy metals can kill marine and fresh water organisms ,disturb biodiversity and harm ecosystems .if acidification is present in water supplies it can damage ecosystem. Aquatic wild life can also suffer from toxic waste as a result of improper e-waste disposal also the air pollution caused by e-waste impact some animal species

- ❖ **Impact on Soil:** One of the most obvious ways that e-waste affects the environment is through the soil. When the improper disposal of e-waste occurs in regular landfills or in places where it is dumped illegally, both heavy metals and flame retardants can leach directly from the e-waste into the soil, causing contamination of underlying

groundwater or contamination of crops that may be planted nearby or in the area in the future. The amount of soil that is contaminated depends on a range of factors, including temperature, soil type, pH levels, and soil composition. When the soil is contaminated by heavy metals, it can be harmful to microorganisms in the soil and plants as a result of exposure to the toxins. Ultimately, animals and wildlife relying on nature for survival will end up consuming affected plants, causing internal health problems.

- ❖ **Impact on Water Quality:** After soil contamination, toxins can eventually make their way into nearby water. Heavy metals from improperly disposed of e-waste, such as mercury, lithium, lead, and barium, can leach through the earth, even reaching the groundwater. When these heavy metals reach groundwater, they eventually make their way into ponds, streams, rivers, and lakes. Local communities often depend on these bodies of water and groundwater. Heavy metal contamination makes clean drinking water for animals and humans problematic to find. Furthermore, when organisms consume this metal, it is stored in trace amounts, accumulates over time, and is then passed up the food chain.
- ❖ **Impact on Human Health:** The environmental impact of e-waste is linked to a rise in a variety of health concerns. Electronic waste

contains toxic components that are dangerous to human health, such as mercury, lead, cadmium, barium, and lithium. As it's in our food and water, exposure to these toxins can pose adverse health effects on humans which include brain, heart, liver, kidney, and skeletal system damage, birth defects (irreversible), human blood contamination as well as central and peripheral nervous systems. These toxins are carcinogenic and can also significantly affect the nervous and reproductive systems of the human body.

- ❖ **Impact on Agriculture:** Agricultural land use is significantly impacted by soil and water pollution. Heavy metals and flame-retardant chemicals can stick around in crops. Livestock, wildlife, and human populations are at increased risk of consuming crops contaminated with these pollutants.
- ❖ **Overflowing Landfill:** Finding effective recycling methods has become a major concern as the global population of electronic devices grows. About 50 million tons of e-waste are generated annually, and only 20% of that is recycled. While most of the remaining 80% ends up in the landfill, knowing full well that they

are non-biodegradable, they accumulate and fill up the landfill over time

- ❖ **Toxic Chemical Release:** Many electronic devices contain hazardous materials like lead, mercury, cadmium, and brominated flame retardants. When improperly disposed off, these chemicals can leach into soil and water, causing pollution and posing health risks to humans and wildlife.
- ❖ **Landfill Contamination:** E-waste often ends up in landfills, taking up space and contributing to the release of harmful substances into the environment. Over time, this can contaminate surrounding soil and groundwater.
- ❖ **Resource Depletion:** Electronics contain valuable resources like metals and minerals. When not recycled, the extraction of these materials from the Earth intensifies, depleting natural resources and causing habitat destruction.
- ❖ **Energy Consumption:** The production, transportation, and disposal of electronics consume significant energy, contributing to greenhouse gas emissions and climate change.

- ❖ Data Security Risks: Improper disposal of electronic devices can lead to data breaches, compromising sensitive information and privacy.
- ❖ Informal Recycling Practices: In some regions, informal recycling methods involve burning or dismantling electronics without safety measures. This not only releases toxins into the air but endangers the health of those involved.

Conclusion

E-waste is not a problem that is going away any time soon. It is only going to get worse. By 2017, the volume of our thrown-away e-products throughout the world is expected to rise by 33 percent from 2012. So, now that we have an idea of the impact of e-waste on the environment, we can dramatically work towards reducing these impacts. Therefore, to avoid these toxic effects of e-waste, it is essential to promote circular economy to reduce environmental harm and promote sustainability.

c) Green computing and cloud computing are two distinct concepts in the field of information technology, with different focuses and objectives:

Green Computing:

Green computing, also known as eco-friendly computing or sustainable computing, is a concept that emphasizes the environmentally responsible use of technology and resources.

Its primary goal is to reduce the environmental impact of computing devices, data centers, and IT operations.

Green computing involves practices such as using energy-efficient hardware, optimizing software to minimize power consumption, recycling electronic waste, and reducing carbon emissions.

The focus is on minimizing the ecological footprint of IT systems and promoting sustainability.

Cloud Computing:

Cloud computing is a technology that provides on-demand access to computing resources and services over the internet, such as servers, storage, databases, and software applications.

Its main objective is to deliver scalable and cost-effective IT solutions by pooling and virtualizing resources, enabling users to access them remotely.

Cloud computing offers benefits like scalability, flexibility, cost-efficiency, and accessibility, making it easier for organizations to manage and deploy their IT infrastructure.

The focus of cloud computing is on improving IT agility and reducing the capital and operational costs associated with traditional on-premises infrastructure.

Conclusion

In summary, green computing focuses on environmentally friendly practices and reducing the environmental impact of IT, while cloud computing focuses on delivering IT services and resources efficiently and cost-effectively over the internet. While they are distinct concepts, they can complement each other, as cloud computing can enable more efficient use of resources, potentially contributing to greener IT practices.

of Cloud Computing.

d) The following are the five benefits of adopting cloud computing:

Cost Efficiency: Cloud computing eliminates the need for expensive upfront hardware and software investments. Users can pay for the resources they use on a pay-as-you-go basis, reducing capital expenses.

Scalability: Cloud services allow businesses to easily scale their computing resources up or down as needed. This flexibility is especially valuable during periods of high demand.

Accessibility: Cloud services are accessible from anywhere with an internet connection, enabling remote work, collaboration, and access to data and applications on the go.

Security and Compliance: Many cloud providers offer robust security measures and compliance certifications, often surpassing what individual businesses can implement on their own. This enhances data protection and regulatory compliance.

Disaster Recovery and Business Continuity: Cloud providers typically have data redundancy and backup systems in place, making it easier for businesses to recover data and maintain operations in case of hardware failures or disasters.

Conclusion.

These advantages make cloud computing an attractive option for businesses looking to improve efficiency, agility, and cost-effectiveness.

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